B.Tech. (Computer Engg.) V Semester Examinations

Session 2022-23

Time: 1 hr. **Computer Networks (CEN - 505) Sessional-II** Max. Marks: 15

Note: Attempt All Questions

1. **CO2**	Consider a source computer (S) transmitting a file of size ~~106~~ 10^6 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2 and L3). L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 100km. Assume signals travel over each link at a speed of ~~108~~ 10^8 meters per second. Assume that the link bandwidth on each link is 1Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D?	4
2. **CO3**	A 12-bit Hamming code whose hexadecimal value is 0*E4F arrives at a receiver. What was the original value in hexadecimal? Assume that not more than 1 bit is in error.	4
3. **CO3**	Show how **Selective-Repeat ARQ** handle the following scenarios: ➢ Lost frame ➢ Lost acknowledgement ➢ Duplicate frame	4
4. **CO3**	A system is generating frame at the rate of 80Mbps, while the rate of the receiver is 8 Mbps for sending acknowledgement. Size of the frame and acknowledgement are 500 bytes and 50 bytes respectively. The round trip delay is 100 μs. Calculate the throughput for stop and wait ARQ.	3